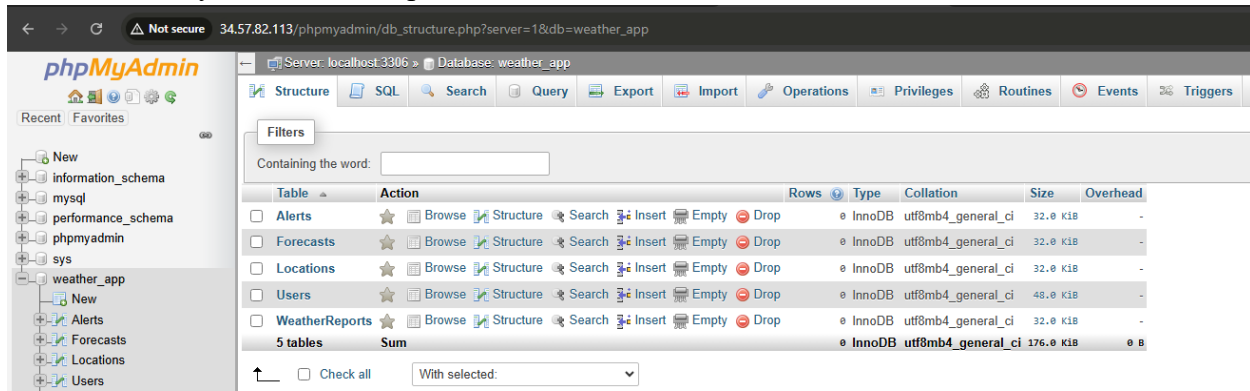


Thomas Hoerger

Project 1

2/24/2025

1. Define your database topic: have at least 5 tables with 30 records for each table:



The screenshot shows the phpMyAdmin interface for a database named 'weather_app'. The left sidebar shows the database structure with 'weather_app' selected. The main panel displays the 'Structure' tab for the 'weather_app' database. It lists 5 tables: Alerts, Forecasts, Locations, Users, and WeatherReports. Each table has 30 records, an InnoDB engine, and a utf8mb4_general_ci collation. The table 'Users' has a size of 48.0 KiB, while the others are 32.0 KiB. A summary row at the bottom shows 5 tables with a total size of 176.0 KiB.

```
INSERT INTO Users (username, email, password) VALUES
('john_doe', 'john@example.com', 'pass123'),
('alice_smith', 'alice@example.com', 'pass456'),
('bob_jones', 'bob@example.com', 'pass789'),
('charlie_adams', 'charlie@example.com', 'pass234'),
('david_lee', 'david@example.com', 'pass567'),
('emma_watson', 'emma@example.com', 'pass890'),
('frank_miller', 'frank@example.com', 'pass345'),
('grace_hill', 'grace@example.com', 'pass678'),
('henry_clark', 'henry@example.com', 'pass901'),
('isabel_roberts', 'isabel@example.com', 'pass012'),
('jason_moore', 'jason@example.com', 'pass123'),
('karen_white', 'karen@example.com', 'pass456'),
('lucas_johnson', 'lucas@example.com', 'pass789'),
('maria_wilson', 'maria@example.com', 'pass234'),
('nathan_brown', 'nathan@example.com', 'pass567'),
('olivia_davis', 'olivia@example.com', 'pass890'),
('paul_anderson', 'paul@example.com', 'pass345'),
('quincy_evans', 'quincy@example.com', 'pass678'),
('rachel_harris', 'rachel@example.com', 'pass901'),
('samuel_clark', 'samuel@example.com', 'pass012'),
('tina_green', 'tina@example.com', 'pass123'),
('ursula_hall', 'ursula@example.com', 'pass456'),
('victor_james', 'victor@example.com', 'pass789'),
('wendy_kim', 'wendy@example.com', 'pass234'),
('xander_lopez', 'xander@example.com', 'pass567'),
('yvonne_martin', 'yvonne@example.com', 'pass890'),
('zack_nelson', 'zack@example.com', 'pass345'),
('leo_perez', 'leo@example.com', 'pass678'),
('mia_quinn', 'mia@example.com', 'pass901'),
('noah_reed', 'noah@example.com', 'pass012');
```

Run SQL query/queries on database weather_app: 

```
1 INSERT INTO Locations (city, country, latitude, longitude) VALUES
2 ('New York', 'USA', 40.7128, -74.0060),
3 ('Los Angeles', 'USA', 34.0522, -118.2437),
4 ('Chicago', 'USA', 41.8781, -87.6298),
5 ('Houston', 'USA', 29.7604, -95.3698),
6 ('Phoenix', 'USA', 33.4484, -112.0740),
7 ('Philadelphia', 'USA', 39.9526, -75.1652),
8 ('San Antonio', 'USA', 29.4241, -98.4936),
9 ('San Diego', 'USA', 32.7157, -117.1611),
10 ('Dallas', 'USA', 32.7767, -96.7970),
11 ('San Jose', 'USA', 37.3382, -121.8863),
12 ('Toronto', 'Canada', 43.6532, -79.3832),
13 ('Vancouver', 'Canada', 49.2827, -123.1207),
14 ('Montreal', 'Canada', 45.5017, -73.5673),
15 ('Mexico City', 'Mexico', 19.4326, -99.1332),
16 ('London', 'UK', 51.5074, -0.1278),
17 ('Manchester', 'UK', 53.4808, -2.2426),
18 ('Paris', 'France', 48.8566, 2.3522),
19 ('Berlin', 'Germany', 52.5200, 13.4050),
20 ('Madrid', 'Spain', 40.4168, -3.7038),
21 ('Rome', 'Italy', 41.9028, 12.4964),
22 ('Tokyo', 'Japan', 35.6895, 139.6917),
23 ('Osaka', 'Japan', 34.6937, 135.5023),
24 ('Seoul', 'South Korea', 37.5665, 126.9780),
25 ('Beijing', 'China', 39.9042, 116.4074),
26 ('Sydney', 'Australia', -33.8688, 151.2093),
27 ('Melbourne', 'Australia', -37.8136, 144.9631),
28 ('Dubai', 'UAE', 25.276987, 55.296249),
29 ('Moscow', 'Russia', 55.7558, 37.6173),
30 ('Minneapolis', 'USA', 44.9778, -93.2650),
31 ('Saint Cloud', 'USA', 45.5602, -94.1621);
```

```

1 INSERT INTO WeatherReports (location_id, temperature, humidity, wind_speed, weather_condition) VALUES
2 (1, 25.3, 60, 12.5, 'Sunny'),
3 (2, 30.1, 55, 8.2, 'Cloudy'),
4 (3, 22.4, 70, 5.6, 'Rainy'),
5 (4, 18.9, 80, 3.3, 'Stormy'),
6 (5, 27.2, 50, 9.8, 'Sunny'),
7 (6, 15.6, 90, 20.0, 'Snowy'),
8 (7, 35.4, 40, 7.5, 'Sunny'),
9 (8, 21.0, 65, 6.3, 'Cloudy'),
10 (9, 19.5, 75, 11.0, 'Rainy'),
11 (10, 24.8, 60, 10.5, 'Sunny'),
12 (11, 28.1, 50, 9.2, 'Clear'),
13 (12, 12.4, 85, 4.8, 'Snowy'),
14 (13, 29.0, 45, 6.7, 'Hot'),
15 (14, 33.6, 30, 5.5, 'Sunny'),
16 (15, 10.2, 92, 3.9, 'Foggy'),
17 (16, 14.5, 88, 4.3, 'Rainy'),
18 (17, 31.2, 35, 8.0, 'Sunny'),
19 (18, 26.8, 60, 6.5, 'Cloudy'),
20 (19, 22.1, 75, 7.1, 'Rainy'),
21 (20, 20.5, 80, 4.4, 'Stormy'),
22 (21, 16.7, 85, 12.0, 'Snowy'),
23 (22, 34.2, 40, 5.8, 'Sunny'),
24 (23, 29.5, 45, 10.1, 'Cloudy'),
25 (24, 18.3, 90, 4.9, 'Rainy'),
26 (25, 24.1, 55, 8.9, 'Clear'),
27 (26, 27.6, 50, 11.3, 'Hot'),
28 (27, 14.8, 93, 3.7, 'Foggy'),
29 (28, 32.4, 35, 9.7, 'Sunny'),
30 (29, 19.0, 70, 6.0, 'Cloudy'),
31 (30, 23.7, 65, 7.4, 'Rainy');

1 INSERT INTO Forecasts (location_id, forecast_date, expected_temp, expected_condition) VALUES
2 (1, '2025-02-25', 27.0, 'Sunny'),
3 (2, '2025-02-26', 31.5, 'Cloudy'),
4 (3, '2025-02-27', 23.0, 'Rainy'),
5 (4, '2025-02-28', 19.5, 'Stormy'),
6 (5, '2025-03-01', 28.0, 'Sunny'),
7 (6, '2025-03-02', 16.0, 'Snowy'),
8 (7, '2025-03-03', 36.0, 'Sunny'),
9 (8, '2025-03-04', 22.0, 'Cloudy'),
10 (9, '2025-03-05', 20.0, 'Rainy'),
11 (10, '2025-03-06', 25.0, 'Sunny'),
12 (11, '2025-03-07', 29.0, 'Clear'),
13 (12, '2025-03-08', 13.0, 'Snowy'),
14 (13, '2025-03-09', 30.0, 'Hot'),
15 (14, '2025-03-10', 34.0, 'Sunny'),
16 (15, '2025-03-11', 11.0, 'Foggy'),
17 (16, '2025-03-12', 15.0, 'Rainy'),
18 (17, '2025-03-13', 32.0, 'Sunny'),
19 (18, '2025-03-14', 27.0, 'Cloudy'),
20 (19, '2025-03-15', 23.0, 'Rainy'),
21 (20, '2025-03-16', 21.0, 'Stormy'),
22 (21, '2025-03-17', 17.0, 'Snowy'),
23 (22, '2025-03-18', 35.0, 'Sunny'),
24 (23, '2025-03-19', 30.0, 'Cloudy'),
25 (24, '2025-03-20', 19.0, 'Rainy'),
26 (25, '2025-03-21', 25.0, 'Clear'),
27 (26, '2025-03-22', 28.0, 'Hot'),
28 (27, '2025-03-23', 15.0, 'Foggy'),
29 (28, '2025-03-24', 33.0, 'Sunny'),
30 (29, '2025-03-25', 20.0, 'Cloudy'),
31 (30, '2025-03-26', 24.0, 'Rainy');

```

```

INSERT INTO Alerts (location_id, alert_message, alert_level, issued_at) VALUES
(1, 'Heat Advisory: Temperatures expected to reach 95°F. Stay hydrated.', 'Severe', '2025-02-20 10:00:00'),
(2, 'Air Quality Alert: High levels of ozone detected. Limit outdoor activities.', 'Moderate', '2025-02-21 09:00:00'),
(3, 'Flood Watch: Heavy rainfall expected. Be prepared for possible flooding.', 'Severe', '2025-02-22 08:30:00'),
(4, 'Hurricane Warning: Category 2 hurricane approaching. Evacuate if advised.', 'Extreme', '2025-02-23 07:45:00'),
(5, 'Excessive Heat Warning: Heat index values up to 110°F.', 'Severe', '2025-02-24 11:00:00'),
(6, 'Winter Storm Watch: Significant snowfall and ice accumulation possible.', 'Moderate', '2025-02-25 12:15:00'),
(7, 'Tornado Watch: Conditions favorable for tornado development.', 'Severe', '2025-02-26 13:00:00'),
(8, 'High Wind Warning: Sustained winds of 40 mph with gusts up to 60 mph.', 'Severe', '2025-02-27 14:30:00'),
(9, 'Flash Flood Warning: Rapid flooding imminent. Seek higher ground immediately.', 'Extreme', '2025-02-28 15:45:00'),
(10, 'Frost Advisory: Temperatures expected to drop to 32°F overnight.', 'Moderate', '2025-03-01 16:00:00'),
(11, 'Dense Fog Advisory: Visibility reduced to 1/4 mile or less.', 'Moderate', '2025-03-02 17:30:00'),
(12, 'Severe Thunderstorm Warning: Expect damaging winds and large hail.', 'Severe', '2025-03-03 18:45:00'),
(13, 'Blizzard Warning: Whiteout conditions expected. Travel will be very dangerous.', 'Extreme', '2025-03-04 19:00:00'),
(14, 'Heat Advisory: High temperatures and humidity expected. Stay cool.', 'Severe', '2025-03-05 20:15:00'),
(15, 'Air Quality Alert: Elevated particulate matter levels. Consider wearing masks.', 'Moderate', '2025-03-06 21:00:00'),
(16, 'Flood Watch: River levels rising. Monitor local forecasts.', 'Severe', '2025-03-07 22:30:00'),
(17, 'Hurricane Watch: Tropical storm strengthening. Prepare for possible hurricane.', 'Extreme', '2025-03-08 23:45:00'),
(18, 'Excessive Heat Warning: Prolonged period of dangerously hot temperatures.', 'Severe', '2025-03-09 06:00:00'),
(19, 'Winter Weather Advisory: Light snow expected. Plan for slippery roads.', 'Moderate', '2025-03-10 07:15:00'),
(20, 'Tornado Warning: Tornado sighted. Take shelter immediately.', 'Extreme', '2025-03-11 08:00:00'),
(21, 'High Wind Watch: Potential for strong winds. Secure loose objects.', 'Severe', '2025-03-12 09:30:00'),
(22, 'Flash Flood Watch: Heavy rain may lead to flash flooding. Stay alert.', 'Severe', '2025-03-13 10:45:00'),
(23, 'Frost Warning: Early-season frost expected. Protect sensitive plants.', 'Moderate', '2025-03-14 11:00:00'),
(24, 'Dense Fog Warning: Visibility near zero in some areas. Use caution.', 'Severe', '2025-03-15 12:30:00'),
(25, 'Severe Thunderstorm Watch: Conditions are ripe for severe storms.', 'Severe', '2025-03-16 13:45:00'),
(26, 'Blizzard Watch: Heavy snow and strong winds possible. Monitor updates.', 'Severe', '2025-03-17 14:00:00'),
(27, 'Heat Warning: Unusually high temperatures expected. Stay indoors if possible.', 'Severe', '2025-03-18 15:15:00'),
(28, 'Air Quality Watch: Potential for poor air quality due to wildfires.', 'Moderate', '2025-03-19 16:00:00'),
(29, 'Flood Advisory: Minor flooding in low-lying areas. Exercise caution.', 'Moderate', '2025-03-20 17:30:00'),
(30, 'Hurricane Advisory: Tropical storm likely to become a hurricane. Stay prepared.', 'Extreme', '2025-03-21 18:45:00');

```

```

UPDATE WeatherReports
SET temperature = (temperature * 9/5) + 32;

```

2. Create a connection with your database: have a query displaying the results on the client side e.g. with html, react-native, QML, or xml:

db_connect.php

```
<?php
$servername = "localhost"; // Change to VPS IP if accessing remotely
$username = "root"; // Use your MySQL username
$password = "Password"; // Use your MySQL root password
$dbname = "weather_app"; // Your database name

// Create connection
$conn = new mysqli($servername, $username, $password, $dbname);

// Check connection
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}
?>
```

weather_display.php

```
<?php
include 'db_connect.php';

// Fetch weather reports with locations
$sql = "SELECT Locations.city, Locations.country, WeatherReports.temperature, WeatherReports.weather_condition
FROM WeatherReports
INNER JOIN Locations USING(location_id)
ORDER BY WeatherReports.temperature ASC";

// Execute query
$result = $conn->query($sql);

// Check for errors
if (!$result) {
    die("Query failed: " . $conn->error);
}
?>

<!DOCTYPE html>
<html lang="en">
<head>
    <title>Weather Reports</title>
    <style>
        body { font-family: Arial, sans-serif; text-align: center; }
        table { width: 80%; margin: auto; border-collapse: collapse; }
        th, td { padding: 10px; border: 1px solid black; }
        th { background-color: #f2a900; color: white; }
        tr:nth-child(even) { background-color: #f9f9f9; }
    </style>
</head>
<body>
    <h2>All Weather Reports Coldest to Warmest</h2>

    <?php if ($result->num_rows > 0): ?>
    <table>
        <tr>
            <th>City</th>
            <th>Country</th>
            <th>Temperature (°C)</th>
            <th>Condition</th>
        </tr>
        <?php while ($row = $result->fetch_assoc()): ?>
        <tr>
            <td><?php echo htmlspecialchars($row['city']); ?></td>
            <td><?php echo htmlspecialchars($row['country']); ?></td>
            <td><?php echo htmlspecialchars($row['temperature']); ?></td>
            <td><?php echo htmlspecialchars($row['weather_condition']); ?></td>
        </tr>
        <?php endwhile; ?>
    </table>
    <?php else: ?>
    <p>No weather reports found.</p>
    <?php endif; ?>
</body>
</html>
```

← → 🔍 Not secure 34.57.82.113/weather_display.php ☆ 📄 🗨 ⋮

All Weather Reports Coldest to Warmest

City	Country	Temperature (°C)	Condition
London	UK	50.36	Foggy
Vancouver	Canada	54.32	Snowy
Manchester	UK	58.1	Rainy
Dubai	UAE	58.64	Foggy
Philadelphia	USA	60.08	Snowy
Tokyo	Japan	62.06	Snowy
Beijing	China	64.94	Rainy
Houston	USA	66.02	Stormy
Minneapolis	USA	66.2	Cloudy
Dallas	USA	67.1	Rainy
Rome	Italy	68.9	Stormy
San Diego	USA	69.8	Cloudy
Madrid	Spain	71.78	Rainy
Chicago	USA	72.32	Rainy
Saint Cloud	USA	74.66	Rainy
Sydney	Australia	75.38	Clear
San Jose	USA	76.64	Sunny
New York	USA	77.54	Sunny
Berlin	Germany	80.24	Cloudy
Phoenix	USA	80.96	Sunny
Melbourne	Australia	81.68	Hot

3. Populate the output table in a presentable form:

← → 🔍 Not secure 34.57.82.113/weather_display.php ☆ 📄 🗨 ⋮

All Weather Reports Coldest to Warmest

City	Country	Temperature (°F)	Condition
London	UK	50.36	Foggy
Vancouver	Canada	54.32	Snowy
Manchester	UK	58.1	Rainy
Dubai	UAE	58.64	Foggy
Philadelphia	USA	60.08	Snowy
Tokyo	Japan	62.06	Snowy
Beijing	China	64.94	Rainy
Houston	USA	66.02	Stormy
Minneapolis	USA	66.2	Cloudy
Dallas	USA	67.1	Rainy
Rome	Italy	68.9	Stormy
San Diego	USA	69.8	Cloudy
Madrid	Spain	71.78	Rainy
Chicago	USA	72.32	Rainy
Saint Cloud	USA	74.66	Rainy
Sydney	Australia	75.38	Clear
San Jose	USA	76.64	Sunny
New York	USA	77.54	Sunny
Berlin	Germany	80.24	Cloudy

New style: (Also changed C to F in the temp collum)

```
<style>
body { font-family: Arial, sans-serif; text-align: center; background-color: #f5f5f5; }
h2 { font-weight: bold; color: black; }
table { width: 80%; margin: auto; border-collapse: collapse; background-color: white; }
th, td { padding: 12px; border: 1px solid black; text-align: center; }
th { background-color: #007BFF; color: white; }
tr:nth-child(even) { background-color: #f9f9f9; }
tr:hover { background-color: #ddd; }
</style>
```


4. Demonstrate the functionality of your database running at least 3 different queries on it:

✓ Showing rows 0 - 4 (5 total, Query took 0.0011 seconds.)

```
SELECT city, country, temperature FROM Locations INNER JOIN WeatherReports USING(location_id) ORDER BY temperature ASC LIMIT 5
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

+ Options

city	country	temperature
London	UK	50.36
Vancouver	Canada	54.32
Manchester	UK	58.1
Dubai	UAE	58.64
Philadelphia	USA	60.08

This retrieves the 5 coldest locations.

```
SELECT COUNT(*) AS RainyDays FROM WeatherReports WHERE weather_condition = 'Rainy'
```

☐ Profiling [Edit inline]

+ Options

RainyDays

6

This counts how many days were recorded as "Rainy".

```
SELECT Locations.city, Locations.country, WeatherReports.wind_speed FROM WeatherReports INNER JOIN Locations USING(location_id) ORDER BY WeatherReports.wind_speed DESC LIMIT 5
```

+ Options

city	country	wind_speed
Philadelphia	USA	20
New York	USA	12.5
Tokyo	Japan	12
Melbourne	Australia	11.3
Dallas	USA	11

Retrieves the 5 windiest locations.

5. Show that you are running your database on your VPS in Linux:

```
tehergs@csci-411-linux:~$ sudo systemctl status mysql
● mysql.service - MySQL Community Server
   Loaded: loaded (/lib/systemd/system/mysql.service; enabled; vendor preset: enabled)
   Active: active (running) since Mon 2025-02-24 21:12:01 UTC; 40min ago
     Process: 2554 ExecStartPre=/usr/share/mysql/mysql-systemd-start pre (code=exited, status=0/SUCCESS)
    Main PID: 2562 (mysqld)
      Status: "Server is operational"
        Tasks: 39 (limit: 1134)
      Memory: 411.0M
      CGroup: /system.slice/mysql.service
              └─2562 /usr/sbin/mysqld

Feb 24 21:11:59 csci-411-linux systemd[1]: Starting MySQL Community Server...
Feb 24 21:12:01 csci-411-linux systemd[1]: Started MySQL Community Server.
tehergs@csci-411-linux:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
   Active: active (running) since Mon 2025-02-24 21:19:27 UTC; 33min ago
     Docs: https://httpd.apache.org/docs/2.4/
     Process: 2673 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
    Main PID: 2677 (apache2)
      Tasks: 11 (limit: 1134)
      Memory: 50.1M
      CGroup: /system.slice/apache2.service
              └─2677 /usr/sbin/apache2 -k start
                  2678 /usr/sbin/apache2 -k start
                  2679 /usr/sbin/apache2 -k start
                  2680 /usr/sbin/apache2 -k start
                  2681 /usr/sbin/apache2 -k start
                  2682 /usr/sbin/apache2 -k start
                  2690 /usr/sbin/apache2 -k start
                  2691 /usr/sbin/apache2 -k start
                  2692 /usr/sbin/apache2 -k start
                  2693 /usr/sbin/apache2 -k start
                  2803 /usr/sbin/apache2 -k start

Feb 24 21:19:27 csci-411-linux systemd[1]: Starting The Apache HTTP Server...
Feb 24 21:19:27 csci-411-linux systemd[1]: Started The Apache HTTP Server.
tehergs@csci-411-linux:~$
```



Not secure 34.57.82.113/weather_display.php

All Weather Reports Coldest to Warmest

City	Country	Temperature (°F)
------	---------	------------------

City	Country	Temperature (°F)
------	---------	------------------

City	Country	Temperature (°F)
------	---------	------------------

3/12/2025

```
tehergs@csci-411-linux:~$ sudo nano /var/www/html/weather_ui.php
```

```
GNU nano 4.8 /var/www/html/weather/ui.php
<?php
/**
 * Weather Database Interactive Viewer
 * Author: Thomas Hoerger - Group 9
 *
 * Description:
 * This program allows users to dynamically select a table from the Weather Database
 * and display its data in the "Display" tab without needing a submit button.
 */

include 'db_connect.php'; // Include database connection file

// Determine which table is selected from the dropdown. Default is "WeatherReports".
$table = isset($_GET['table']) ? $_GET['table'] : 'WeatherReports';

$displayName = preg_replace('/(?<!\\ ) [A-Z]/', ' $0', $table); // Add spaces before uppercase letters
$displayName = ucwords(str_replace('_', ' ', trim($displayName))); // Format and capitalize

// Fetch the first 30 rows from the selected table
$sql = "SELECT * FROM $table LIMIT 30";
$result = $conn->query($sql);

// Retrieve column headers dynamically from the selected table
$headers = [];
if ($result && $result->num_rows > 0) {
    $headers = array_keys($result->fetch_assoc()); // Extract column names from the first row
    $result->data_seek(0); // Reset result pointer to fetch rows again
}
?>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="author" content="Thomas Hoerger - Group 9">
    <meta name="description" content="Interactive Weather Database Viewer">
    <title>Weather Database - Group 9</title>

    <style>
        /* General styling for the webpage */
        body {
            font-family: Arial, sans-serif;
            background-color: #f5f5f5;
            text-align: center;
        }

        /* Title styling */
        h1 {
            font-weight: bold;
            color: black;
        }
    </style>

```


2nd part of code:

```
/* Navigation tab container */
.tab-container {
  display: flex;
  justify-content: center;
  margin: 20px 0;
}

/* Styling for each tab button */
.tab {
  padding: 12px 20px;
  cursor: pointer;
  background-color: #007BFF;
  color: white;
  border: 1px solid black;
  margin: 0 5px;
}

/* Change tab color on hover */
.tab:hover {
  background-color: #0056b3;
}

/* Highlight the active tab */
.tab.active {
  background-color: #003f7f;
}

/* Hide all content sections by default */
.content {
  display: none;
  margin-top: 20px;
}

/* Show the active content section */
.content.active {
  display: block;
}

/* Styling for tables */
table {
  width: 80%;
  margin: auto;
  border-collapse: collapse;
  background-color: white;
}

/* Styling for table headers and data cells */
th, td {
  padding: 12px;
  border: 1px solid black;
  text-align: center;
}
```

3rd part of code:

```
/* Styling for table headers */
th {
    background-color: #007BFF;
    color: white;
}

/* Alternate row coloring for readability */
tr:nth-child(even) {
    background-color: #f9f9f9;
}

/* Highlight rows when hovered */
tr:hover {
    background-color: #ddd;
}
</style>
</head>
<body>

<!-- Main Title -->
<h1>Weather Database Viewer</h1>
<p><strong>Author:</strong> Thomas Hoerger - Group 9</p>

<!-- Navigation Tabs -->
<div class="tab-container">
    <div class="tab active" onclick="showTab('selectTab')">Select</div>
    <div class="tab" onclick="showTab('displayTab')">Display</div>
</div>

<!-- Select Table Section -->
<div id="selectTab" class="content active">
    <h2>Select a Table</h2>

    <!-- Dropdown menu to select a table (auto-updates on selection change) -->
    <label for="table">Choose a Table:</label>
    <select name="table" id="table" onchange="changeTable(this.value)">
        <option value="WeatherReports" <?=( $table == 'WeatherReports' ) ? 'selected' : '' ?>>Weather Reports</option>
        <option value="Locations" <?=( $table == 'Locations' ) ? 'selected' : '' ?>>Locations</option>
        <option value="Forecasts" <?=( $table == 'Forecasts' ) ? 'selected' : '' ?>>Forecasts</option>
    </select>
</div>

<!-- Display Table Section -->
<div id="displayTab" class="content">
    <h2><?=( htmlspecialchars($displayTableName) ) ?></h2> <!-- Display selected table name -->

    <!-- Display table only if data exists -->
    <?php if ( $result && $result->num_rows > 0 ): ?>
    <table>
        <tr>
            <?php
                // Generate table headers dynamically
                foreach ( $headers as $header ): ?>
                    <th><?=( htmlspecialchars($header) ); ?></th>
                <?php endforeach; ?>
            </tr>
        </table>
    </div>
```

4th part of code:

```
</tr>
<?php endwhile; ?>
</table>
<?php else: ?>
    <!-- Display message if no data is available -->
    <p>No data available in this table.</p>
<?php endif; ?>
</div>

<script>
/**
 * Function to switch between tabs dynamically
 * @param {string} tabId - The ID of the tab to be displayed
 */
function showTab(tabId) {
    // Hide all tab content sections
    document.querySelectorAll('.content').forEach(tab => tab.classList.remove('active'));
    document.querySelector('.tab').forEach(t => t.classList.remove('active'));

    // Show the selected tab's content
    document.getElementById(tabId).classList.add('active');

    // Mark the selected tab as active
    document.querySelector(`.tab[onclick="showTab('${tabId}')"`).classList.add('active');
}

/**
 * Function to update the displayed table automatically when a new one is selected
 * @param {string} tableName - The selected table name
 */
function changeTable(tableName) {
    window.location.href = "?table=" + tableName; // Update the URL to refresh data dynamically
}
</script>

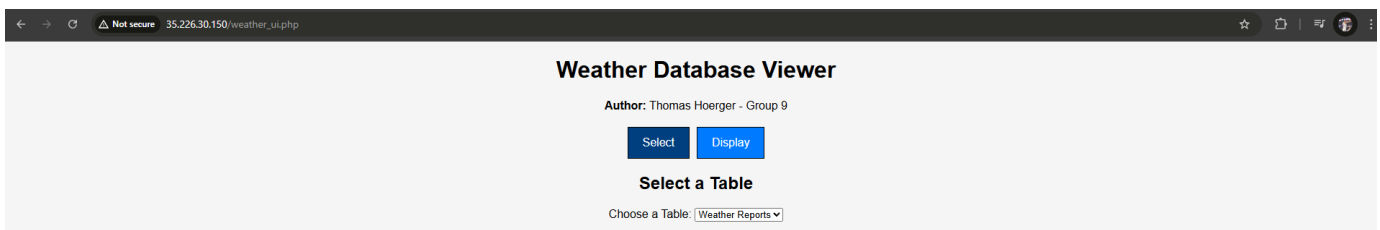
</body>
</html>

<?php
// Close the database connection
$conn->close();
?>
```

Step 3 restart Apache to apply changes:

```
tehergs@csci-411-linux:~$ sudo systemctl restart apache2
```

Step 4 go to [http://\(External IP\)/weather_ui.php](http://(External IP)/weather_ui.php):



Weather Database Viewer

Author: Thomas Hoerger - Group 9

Select

Display

Weather Reports

report_id	location_id	temperature	humidity	wind_speed	weather_condition	report_time
1	1	77.54	60	12.5	Sunny	2025-02-24 20:25:37
2	2	86.18	55	8.2	Cloudy	2025-02-24 20:25:37
3	3	72.32	70	5.6	Rainy	2025-02-24 20:25:37
4	4	66.02	80	3.3	Stormy	2025-02-24 20:25:37
5	5	80.96	50	9.8	Sunny	2025-02-24 20:25:37
6	6	60.08	90	20	Snowy	2025-02-24 20:25:37
7	7	95.72	40	7.5	Sunny	2025-02-24 20:25:37
8	8	69.8	65	6.3	Cloudy	2025-02-24 20:25:37
9	9	67.1	75	11	Rainy	2025-02-24 20:25:37
10	10	76.64	60	10.5	Sunny	2025-02-24 20:25:37
11	11	82.58	50	9.2	Clear	2025-02-24 20:25:37
12	12	54.32	85	4.8	Snowy	2025-02-24 20:25:37
13	13	84.2	45	6.7	Hot	2025-02-24 20:25:37
14	14	92.48	30	5.5	Sunny	2025-02-24 20:25:37
15	15	50.36	92	3.9	Foggy	2025-02-24 20:25:37

Weather Database Viewer

Author: Thomas Hoerger - Group 9

Select

Display

Locations

location_id	user_id	city	country	latitude	longitude
1		New York	USA	40.71280000	-74.00600000
2		Los Angeles	USA	34.05220000	-118.24370000
3		Chicago	USA	41.87810000	-87.62980000
4		Houston	USA	29.76040000	-95.36980000
5		Phoenix	USA	33.44840000	-112.07400000
6		Philadelphia	USA	39.95260000	-75.16520000
7		San Antonio	USA	29.42410000	-98.49360000
8		San Diego	USA	32.71570000	-117.16110000
9		Dallas	USA	32.77670000	-96.79700000
10		San Jose	USA	37.33820000	-121.88630000

Weather Database Viewer

Author: Thomas Hoerger - Group 9

Select

Display

Forecasts

forecast_id	location_id	forecast_date	expected_temp	expected_condition
1	1	2025-02-25	27	Sunny
2	2	2025-02-26	31.5	Cloudy
3	3	2025-02-27	23	Rainy
4	4	2025-02-28	19.5	Stormy
5	5	2025-03-01	28	Sunny
6	6	2025-03-02	16	Snowy
7	7	2025-03-03	36	Sunny
8	8	2025-03-04	22	Cloudy
9	9	2025-03-05	20	Rainy
10	10	2025-03-06	25	Sunny
11	11	2025-03-07	29	Clear
12	12	2025-03-08	13	Snowy
13	13	2025-03-09	30	Hot
14	14	2025-03-10	34	Sunny
15	15	2025-03-11	11	Foggy
16	16	2025-03-12	15	Rainy

23	23	2025-03-19	30	Cloudy
24	24	2025-03-20	19	Rainy
25	25	2025-03-21	25	Clear
26	26	2025-03-22	28	Hot
27	27	2025-03-23	15	Foggy
28	28	2025-03-24	33	Sunny
29	29	2025-03-25	20	Cloudy
30	30	2025-03-26	24	Rainy

Step 5 show MySQL is running:

```
tehergs@csci-411-linux:~$ sudo systemctl status mysql
● mysql.service - MySQL Community Server
   Loaded: loaded (/lib/systemd/system/mysql.service; enabled; vendor preset: enabled)
   Active: active (running) since Thu 2025-03-13 01:21:25 UTC; 28min ago
     Main PID: 627 (mysqld)
    Status: "Server is operational"
      Tasks: 39 (limit: 1134)
     Memory: 407.6M
    CGroup: /system.slice/mysql.service
            └─627 /usr/sbin/mysqld

Mar 13 01:21:16 csci-411-linux systemd[1]: Starting MySQL Community Server...
Mar 13 01:21:25 csci-411-linux systemd[1]: Started MySQL Community Server.
tehergs@csci-411-linux:~$
```

Step 6 show Apache is running:

```
tehergs@csci-411-linux:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
   Active: active (running) since Thu 2025-03-13 01:46:21 UTC; 4min 42s ago
     Docs: https://httpd.apache.org/docs/2.4/
  Process: 2274 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
     Main PID: 2278 (apache2)
      Tasks: 7 (limit: 1134)
     Memory: 39.4M
    CGroup: /system.slice/apache2.service
            └─2278 /usr/sbin/apache2 -k start
              └─2279 /usr/sbin/apache2 -k start
                └─2280 /usr/sbin/apache2 -k start
                  └─2281 /usr/sbin/apache2 -k start
                    └─2282 /usr/sbin/apache2 -k start
                      └─2283 /usr/sbin/apache2 -k start
                        └─2284 /usr/sbin/apache2 -k start

Mar 13 01:46:21 csci-411-linux systemd[1]: Starting The Apache HTTP Server...
Mar 13 01:46:21 csci-411-linux systemd[1]: Started The Apache HTTP Server.
tehergs@csci-411-linux:~$
```

Thomas Hoerger

Project 2

3/27/2025

1. Create a Login Page:

```
tehergs@csci-411-linux:~$ sudo nano /var/www/html/login.php
```

2. Write Login Page Code:

```
GNU nano 4.8 /var/www/html/login.php
<!DOCTYPE html>
<html>
<head>
  <title>Skyview Login</title>
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <style>
    body {
      font-family: Arial, sans-serif;
      background: #e0f7fa;
      margin: 0;
      padding: 0;
    }

    .login-box {
      max-width: 400px;
      margin: 10vh auto;
      background: white;
      padding: 20px;
      border-radius: 8px;
      box-shadow: 0px 4px 10px rgba(0,0,0,0.1);
    }

    input, button {
      width: 100%;
      padding: 10px;
      margin: 8px 0;
      font-size: 16px;
      border-radius: 5px;
      border: 1px solid #ccc;
    }

    button {
      background-color: #007BFF;
      color: white;
      border: none;
    }

    button:hover {
      background-color: #0056b3;
    }

    @media screen and (max-width: 480px) {
      .login-box {
        margin: 5vh 10px;
      }
    }
  </style>
</head>
<body>
  <div class="login-box">
    <h2>Login to Skyview Weather</h2>
    <form method="POST" action="auth.php">
      <input type="email" name="email" placeholder="Email" required><br>
      <input type="password" name="password" placeholder="Password" required><br>
      <button type="submit">Login</button>
    </form>
  </div>
  <p style="margin: 15px 0; font-weight: bold; text-align: center;">— OR —</p>
  <a href="google-login.php"><button>Login with Google</button></a>
</body>
</html>
```

3. Create auth.php to Handle Logins:

```
tehergs@csci-411-linux:~$ sudo nano /var/www/html/auth.php
```

4. Write Code for auth.php:

```
GNU nano 4.8 /var/www/html/auth.php Modified
<?php
session_start();
include 'db_connect.php'; // Reuse your existing DB connection

// Get email and password from the form submission
$email = $_POST['email'];
$password = $_POST['password'];

// In production, hash and verify passwords securely
$sql = "SELECT * FROM Users WHERE email = '$email' AND password = '$password'";
$result = $conn->query($sql);

// Check if login is valid
if ($result->num_rows > 0) {
    $row = $result->fetch_assoc();
    $_SESSION['user_email'] = $row['email'];
    $_SESSION['user_name'] = $row['username'];
    header("Location: weather_ui.php");
} else {
    echo "Invalid login. <a href='login.php'>Try again</a>";
}
?>
```

5. Set Up Google OAuth 2.0 Login:

OAuth 2.0 Client IDs

☐	Name	Creation date ↓	Type	Client ID	Actions
☐	WeatherLoginClient	Mar 27, 2025	Web application	913995445558-eop3...	

Google Auth Platform / Clients / Client: 913995445558-eop3jnicnggj46n14mgfheqo33jlfqdh.apps.googleusercontent.com

← Client ID for Web application Delete

Name *
WeatherLoginClient
The name of your OAuth 2.0 client. This name is only used to identify the client in the console and will not be shown to end users.

The domains of the URIs you add below will be automatically added to your [OAuth consent screen](#) as [authorized domains](#).

Authorized JavaScript origins ②

For use with requests from a browser

+ Add URI

Authorized redirect URIs ②

For use with requests from a web server

URIs 1 *
<http://skyview.work.gd/google-callback.php>

+ Add URI

Note: It may take 5 minutes to a few hours for settings to take effect

Save Cancel

Additional information

Client ID	913995445558-eop3jnicnggj46n14mgfheqo33jlfqdh.apps.googleusercontent.com
Creation date	March 27, 2025 at 6:48:59 PM GMT-5

Client secrets

If you are in the process of changing client secrets, you can manually rotate them without downtime. [Learn more](#)

Client secret	GOCSPX-IoXf5sJjnr4NJaZ2Ck5b4Gnc4Ql
Creation date	March 27, 2025 at 6:48:59 PM GMT-5
Status	Enabled

+ Add secret

6. Install Google OAuth PHP Library:

```
tehergs@csci-411-linux:~$ sudo apt install composer
```

```
tehergs@csci-411-linux:/var/www/html$ sudo composer require google/apiclient:^2.0
Do not run Composer as root/super user! See https://getcomposer.org/root for details
Warning from https://repo.packagist.org: Support for Composer 1 will be shutdown on August 1st 2025. You should
upgrade to Composer 2. See https://blog.packagist.com/shutting-down-packagist-org-support-for-composer-1-x/
./composer.json has been updated
Loading composer repositories with package information
Warning from https://repo.packagist.org: Support for Composer 1 will be shutdown on August 1st 2025. You should
upgrade to Composer 2. See https://blog.packagist.com/shutting-down-packagist-org-support-for-composer-1-x/
Updating dependencies (including require-dev)
Package operations: 18 installs, 0 updates, 0 removals
 - Installing ralouphie/getallheaders (3.0.3): Downloading (100%)
 - Installing psr/http-message (2.0): Downloading (100%)
 - Installing psr/http-factory (1.1.0): Downloading (100%)
 - Installing guzzlehttp/psr7 (2.7.0): Downloading (100%)
 - Installing symfony/deprecation-contracts (v2.5.4): Downloading (100%)
 - Installing psr/http-client (1.0.3): Downloading (100%)
 - Installing guzzlehttp/promises (2.0.4): Downloading (100%)
 - Installing guzzlehttp/guzzle (7.9.2): Downloading (100%)
 - Installing paragonie/random_compat (v9.99.100): Downloading (100%)
 - Installing paragonie/constant_time_encoding (v2.7.0): Downloading (100%)
 - Installing phpseclib/phpseclib (3.0.43): Downloading (100%)
 - Installing psr/log (1.1.4): Downloading (100%)
 - Installing monolog/monolog (2.10.0): Downloading (100%)
 - Installing firebase/php-jwt (v6.10.0): Downloading (100%)
 - Installing google/apiclient-services (v0.355.0): Downloading (100%)
 - Installing psr/cache (1.0.1): Downloading (100%)
 - Installing google/auth (v1.37.2): Downloading (100%)
 - Installing google/apiclient (v2.16.1): Downloading (100%)
guzzlehttp/psr7 suggests installing laminas/laminas-httphandler (Emit PSR-7 responses)
guzzlehttp/guzzle suggests installing ext-intl (Required for Internationalized Domain Name (IDN) support)
paragonie/random_compat suggests installing ext-libsodium (Provides a modern crypto API that can be used to gene
rate random bytes.)
phpseclib/phpseclib suggests installing ext-libsodium (SSH2/SFTP can make use of some algorithms provided by the
```

7. Create google-callback.php (Handles Google Login Response):

```
tehergs@csci-411-linux:~$ sudo nano /var/www/html/google-callback.php
```

```
GNU nano 4.8 /var/www/html/google-callback.php Modifi
<?php
require_once 'vendor/autoload.php';
session_start();

$client = new Google_Client();
$client->setClientId('913995445558-eop3jnicnggj46n14mgfhego33jlfqdh.apps.googleusercontent.com');
$client->setClientSecret('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');
$client->setRedirectUri('http://skyview.work.gd/google-callback.php'); // MUST match Google OAuth settings
$client->addScope("email");
$client->addScope("profile");

if (isset($_GET['code'])) {
    $token = $client->fetchAccessTokenWithAuthCode($_GET['code']);
    $client->setAccessToken($token);

    // Get user info from Google
    $oauth = new Google_Service_Oauth2($client);
    $user = $oauth->userinfo->get();

    // Store user info in session
    $_SESSION['user_email'] = $user->email;
    $_SESSION['user_name'] = $user->name;

    // Redirect to your weather app UI
    header('Location: weather_ui.php');
    exit();
} else {
    echo "Error: No authorization code received.";
}
?>
```

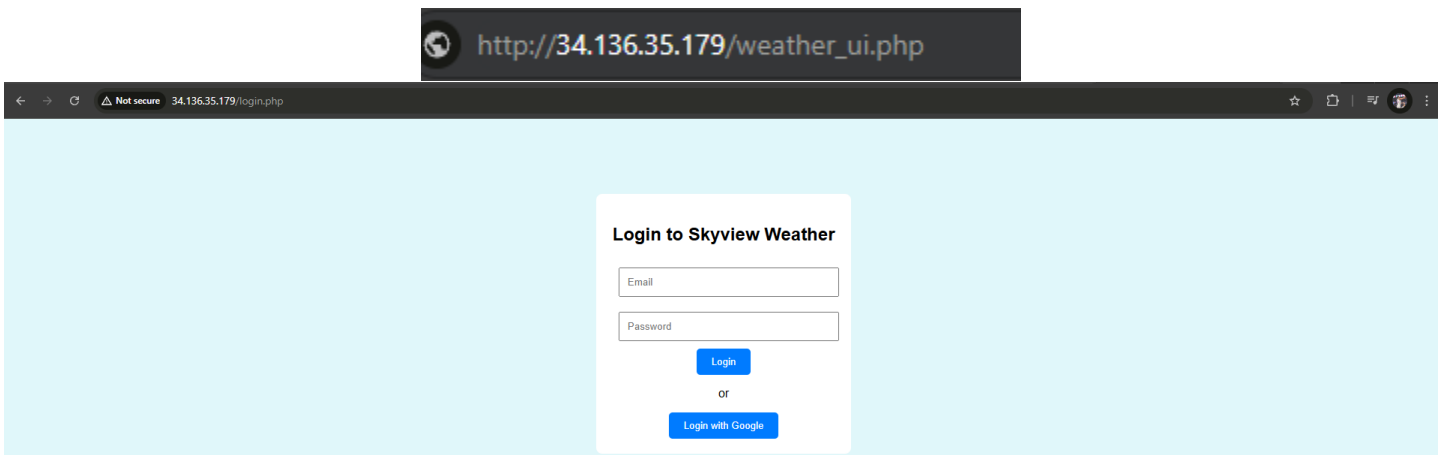
8. Update weather_ui.php to Require Login:

```
GNU nano 4.8 /var/www/html/weather_ui.php
<?php
session_start(); // Add this line first to enable session tracking

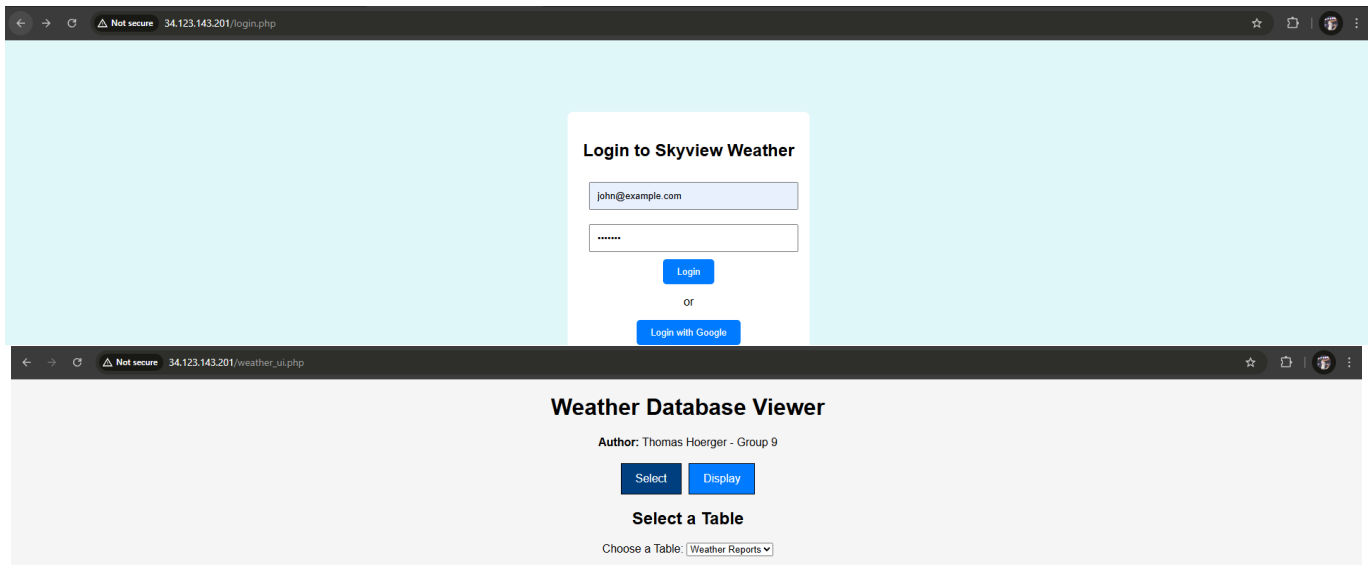
// Redirect user to login page if they are not logged in
if (!isset($_SESSION['user_email']) && !isset($_SESSION['user'])) {
    header("Location: login.php");
    exit();
}

/**
 * Weather Database Interactive Viewer
 * Author: Thomas Hoerger - Group 9
 *
 * Description:
 * This program allows users to dynamically select a table from the Weather Database
 * and display its data in the "Display" tab without needing a submit button.
 */
```

9. Test to Make Sure it Redirects You to login.php if You're Not Logged in:



10. Test a Test User:



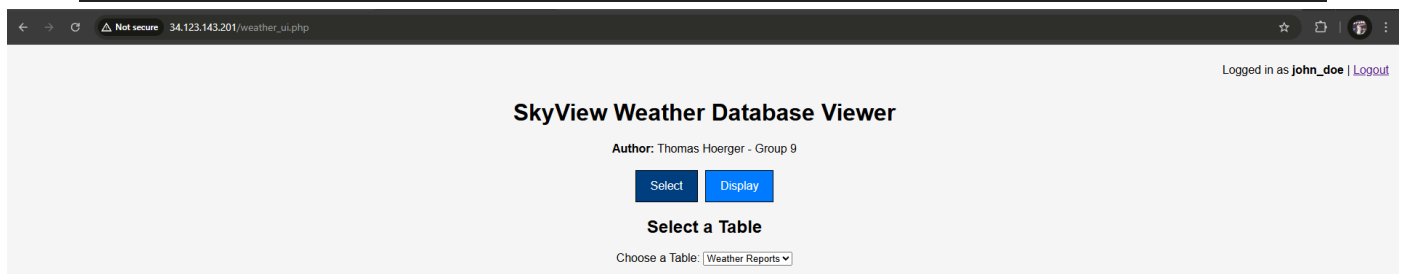
11. Create a Logout Option:

```
tehergs@csci-411-linux:~$ sudo nano /var/www/html/logout.php

GNU nano 4.8 /var/www/html/logout.php
<?php
session_start();           // Start the current session
session_destroy();         // Destroy all session data
header("Location: login.php"); // Redirect back to login page
exit();
?>
```

12. Add a Logout Link to weather_ui.php:

```
<p style="text-align: right; padding: 10px;">
  Logged in as <strong><?= htmlspecialchars($_SESSION['user_name']) ?? $_SESSION['user_email'] ?? 'User' ?></strong> |
  <a href="logout.php">Logout</a>
</p>
```



13. Have your HCI/GUI/UI/UX running on at least 2 platforms: laptop and mobile (Edit Style Sheet):

```
GNU nano 4.8 /var/www/html/weather_ui.php
<meta name="viewport" content="width=device-width, initial-scale=1.0">
GNU nano 4.8 /var/www/html/login.php
<!DOCTYPE html>
<html>
<head>
<title>Skyview Login</title>
<meta name="viewport" content="width=device-width, initial-scale=1">
<style>
  body {
    font-family: Arial, sans-serif;
    background: #e0f7fa;
    margin: 0;
    padding: 0;
  }

  .login-box {
    max-width: 400px;
    margin: 10vh auto;
    background: white;
    padding: 20px;
    border-radius: 8px;
    box-shadow: 0px 4px 10px rgba(0,0,0,0.1);
  }

  input, button {
    width: 100%;
    padding: 10px;
    margin: 8px 0;
    font-size: 16px;
    border-radius: 5px;
    border: 1px solid #ccc;
  }

  button {
    background-color: #007BFF;
    color: white;
    border: none;
  }

  button:hover {
    background-color: #0056b3;
  }

  @media screen and (max-width: 480px) {
    .login-box {
      margin: 5vh 10px;
    }
  }
</style>
```

```
<title>Weather Database - Group 9</title>
<style>
  body {
    font-family: Arial, sans-serif;
    text-align: center;
    background-color: #f5f5f5;
    margin: 0;
    padding: 0;
  }

  h1, h2 {
    font-weight: bold;
    color: black;
    margin: 10px;
  }

  .tab-container {
    display: flex;
    flex-wrap: wrap;
    justify-content: center;
    margin: 10px;
  }

  .tab {
    padding: 12px 20px;
    cursor: pointer;
    background-color: #007bff;
    color: white;
    border: 1px solid black;
    margin: 5px;
    border-radius: 6px;
  }

  .tab.active {
    background-color: #003f7f;
  }

  .content {
    display: none;
    margin: 10px auto;
    width: 95%;
  }

  .content.active {
    display: block;
  }

  table {
    width: 100%;
    border-collapse: collapse;
    background-color: white;
    margin-top: 15px;
    font-size: 14px;
  }

  th, td {
    padding: 10px;
    border: 1px solid #ccc;
    text-align: center;
  }
}
```

```

th {
    background-color: #007BFF;
    color: white;
}

tr:nth-child(even) {
    background-color: #f9f9f9;
}

tr:hover {
    background-color: #ddd;
}

select {
    padding: 8px;
    font-size: 16px;
    margin-top: 10px;
    width: 90%;
    max-width: 300px;
}

@media screen and (max-width: 600px) {
    .tab {
        width: 90%;
        font-size: 16px;
    }

    table, th, td {
        font-size: 12px;
        padding: 8px;
    }
}
</style>

```

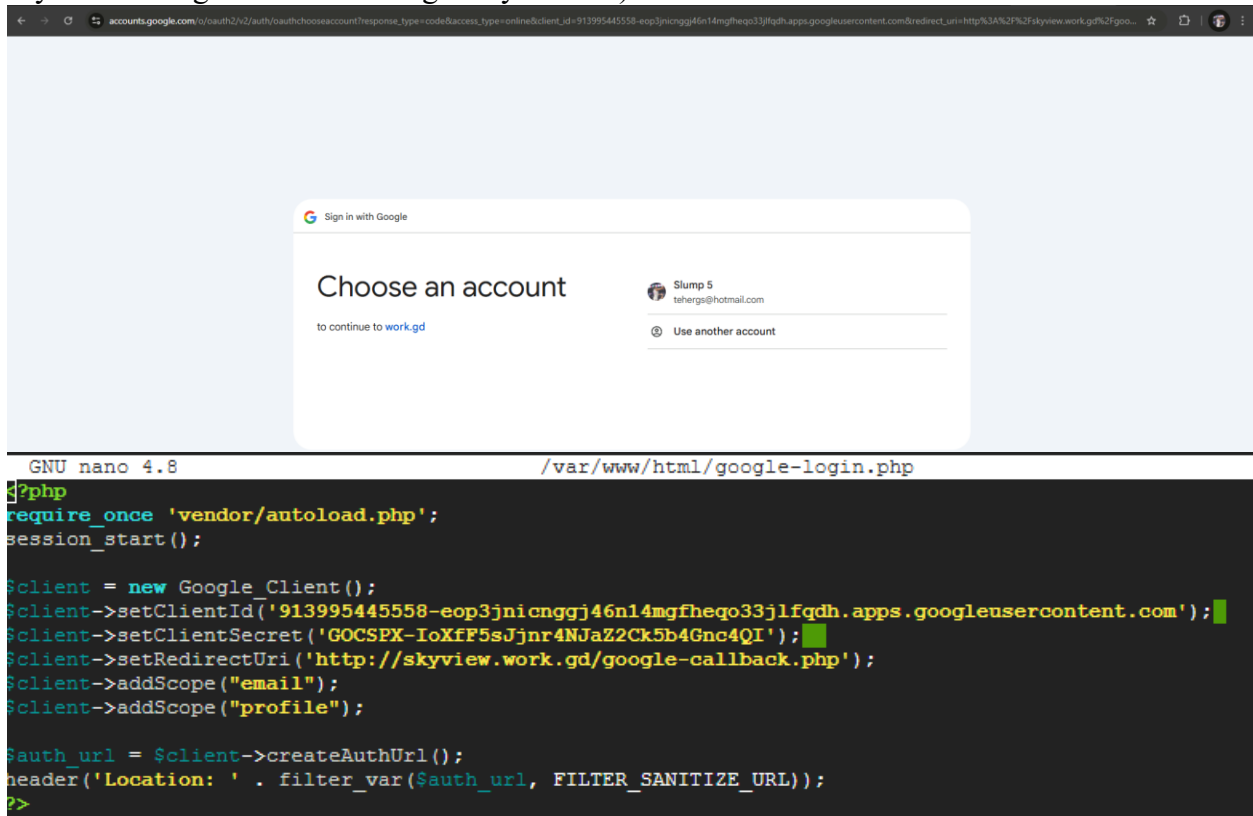
```

<div class="table-wrapper">
    <table>
        <tr>
            <?php foreach ($headers as $header): ?>
                <th><?= htmlspecialchars($header); ?></th>
            <?php endforeach; ?>
        </tr>
        <?php while ($row = $result->fetch_assoc()): ?>
            <tr>
                <?php foreach ($headers as $header): ?>
                    <td><?= htmlspecialchars($row[$header]); ?></td>
                <?php endforeach; ?>
            </tr>
        <?php endwhile; ?>
    </table>
</div>

```

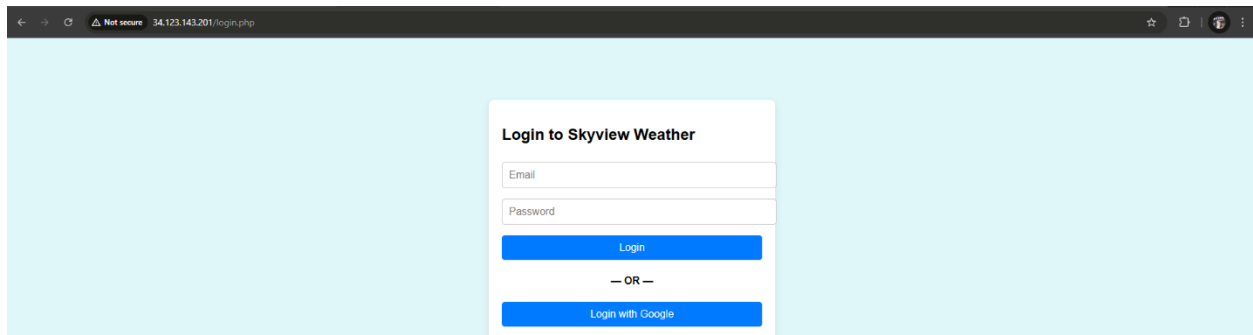
```
.table-wrapper {
    width: 100%;
    overflow-x: auto;
    display: flex;
    justify-content: center;
    margin-top: 10px;
}
```

14. Test Google Login (Works just need to wait for DNS propagation delay to use skyview.work.gd because I changed my VPS IP)

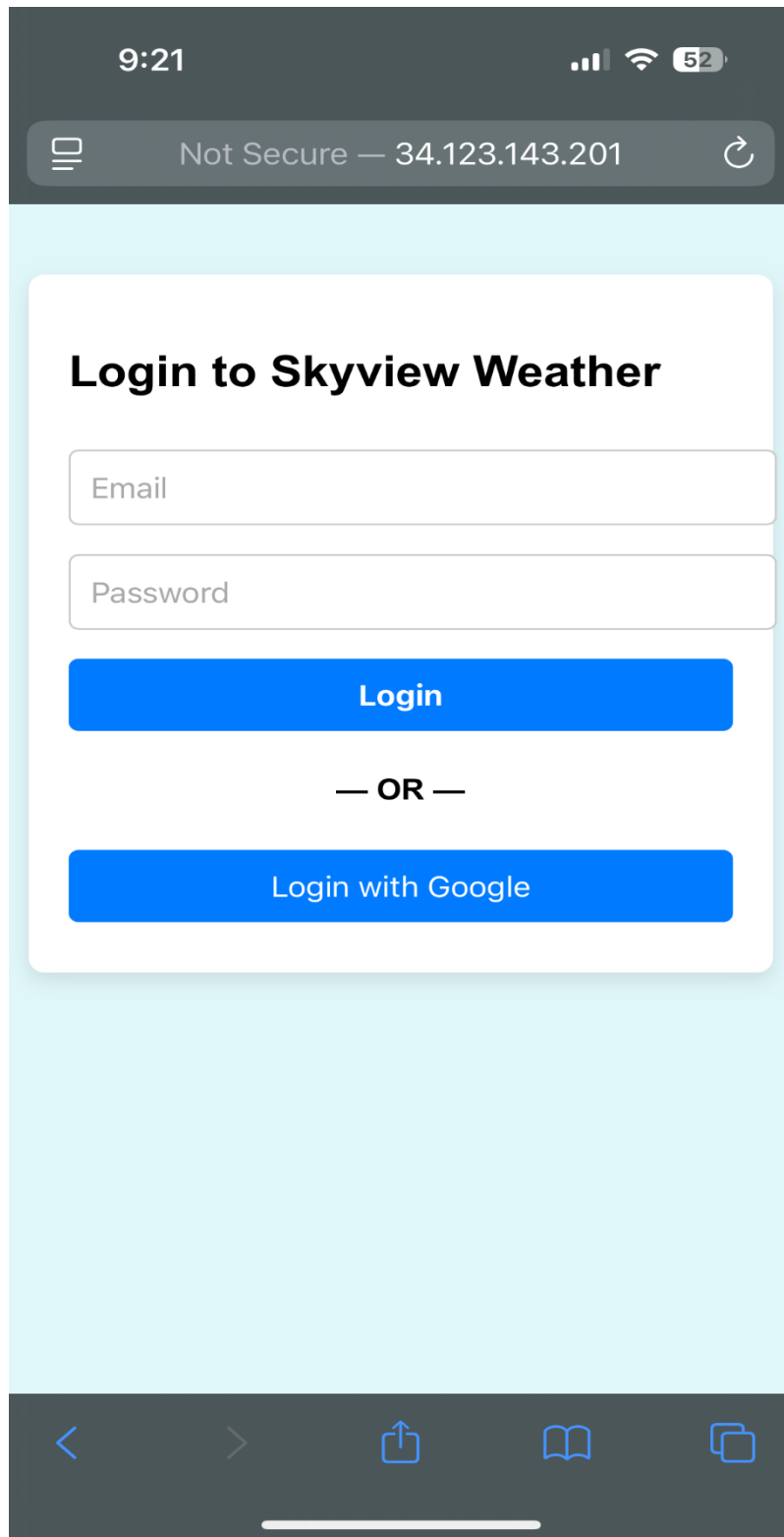


15. Test Website on PC and Mobile

PC:



Mobile:



A mobile app login screen for "Skyview Weather". The screen has a light blue background. At the top, there is a dark grey status bar with the time "9:21", signal strength, Wi-Fi, and battery level "52". Below the status bar is a dark grey browser-like header with a menu icon, the text "Not Secure — 34.123.143.201", and a refresh icon. The main content is a white rounded rectangle with a shadow. Inside, the title "Login to Skyview Weather" is in bold black text. Below the title are two input fields: "Email" and "Password". Under the "Password" field is a blue "Login" button. Below the button is the text "— OR —". At the bottom of the white rectangle is a blue button labeled "Login with Google". At the very bottom of the screen is a dark grey navigation bar with five icons: a back arrow, a forward arrow, a share icon, a bookmark icon, and a tabs icon. A white horizontal line is visible at the bottom of the screen, likely representing the home indicator.

9:21 52

Not Secure — 34.123.143.201

Login to Skyview Weather

Email

Password

Login

— OR —

Login with Google

